Data Science & Data Analysis Workflow Guide

The Complete Industry Roadmap — From Raw Data to Business Decisions

A structured, phase-by-phase breakdown of the real-world data workflow used inside companies — built so you can see exactly what to learn, why it matters on the job, and how it fits into the bigger picture of an industry data project.

Why This Field Is Booming

Every industry — retail, banking, healthcare, logistics, SaaS — now runs on data. Companies don't just want people who can build models; they want people who can turn messy data into decisions. That's why Data Analysis and Data Science remain two of the fastest-growing, most hireable skill sets: the demand isn't just for AI researchers, it's for people who can clean data, find patterns, explain them clearly, and help a business act on them.

How the Industry Workflow Actually Works

In a real company, a data project rarely starts with a fancy model — it starts with a business question. The typical flow looks like this:

Business Problem	Data Collection	Data Cleaning	EDA (Exploratory Analysis)	Statistics & Hypothesis Testing	Modeling / ML	Evaluation	Deployment & Reporting	Business Decision

This guide is organized around that same flow — each phase below tells you what industry actually expects at that stage, plus exactly where to learn it.

Where to Learn It

YouTube Channels & Learning Platforms for the Data Science / Data Analysis Roadmap

Companion reference to the Data Science Industry Workflow above. A quick-reference list of where to actually learn each phase — organized so you can pick a resource and start immediately instead of searching around.

Phase	What It's Actually For (Industry Use)	YouTube Channels / Videos	Platforms / Courses
Phase 1 Foundations (Python, Excel, SQL, Statistics)	Every analyst/DS role starts here — pulling data from databases, cleaning it in Python/Excel, and knowing basic stats to avoid wrong conclusions.	• freeCodeCamp — full Python course • Corey Schafer — Python + Pandas playlist • freeCodeCamp — "SQL Full Course" • Alex The Analyst — Data Analyst full roadmap • StatQuest — statistics fundamentals	• Kaggle Learn — free Python/SQL/Pandas micro-courses • Google Data Analytics Certificate (Coursera) • Khan Academy — statistics
Phase 2 Data Collection & Cleaning (ETL)	Real company data is messy — 60-80% of on-the-job time is spent here: handling missing values, duplicates, inconsistent formats, and merging sources.	• Alex The Analyst — data cleaning in Excel/SQL • Rob Mulla — pandas data cleaning • freeCodeCamp — ETL/Airflow basics	• Kaggle — real messy datasets to practice on • DataCamp — Data Cleaning in Python (free preview) • dbt Labs Learn — modern ETL/ELT practices
Phase 3 Exploratory Data Analysis (EDA) & Visualization	This is how analysts "talk" to data before modeling — spotting trends, outliers, and patterns, then communicating them through charts dashboards teams actually use.	• Alex The Analyst — Power BI / Tableau playlists • Ken Jee — data science project walkthroughs • Luke Barousse — data analyst portfolio projects	• Kaggle Learn — Data Visualization course • Tableau Public — free training • Microsoft Learn — Power BI path
Phase 4 Statistics & Hypothesis Testing	Separates a real analyst from someone who just makes charts — A/B testing, confidence intervals, and correlation-vs-causation come up constantly in product and business decisions.	• StatQuest with Josh Starmer — hypothesis testing, A/B tests • Krish Naik — statistics for data science	• Khan Academy — inferential statistics • Udacity — "Intro to Statistics" (free) • DeepLearning.AI — short course on A/B testing
Phase 5 Machine Learning for Analytics	Used for forecasting, churn prediction, segmentation, and recommendation — the ML that actually ships in most companies is simpler than research-lab ML.	• StatQuest — ML concepts explained simply • Krish Naik — Machine Learning playlist • Ken Jee — applied ML case studies	• Andrew Ng's Machine Learning Specialization (Coursera/DeepLearning.AI — financial aid available) • Kaggle competitions for hands-on practice
Phase 6 Business Intelligence & Dashboarding	The deliverable most stakeholders actually see — a live dashboard, not a Jupyter notebook. Fluency in BI tools is a top hiring requirement for analyst roles.	• Alex The Analyst — Power BI full course • Pragmatic Works — Power BI/SSRS tutorials • Tableau — official YouTube channel	• Microsoft Learn — Power BI learning path • Tableau eLearning (free modules) • Google — Looker Studio help center

Phase	What It's Actually For (Industry Use)	YouTube Channels / Videos	Platforms / Courses
Phase 7 Big Data & Cloud Tools	Once data outgrows a laptop, companies move to cloud warehouses and distributed processing — SQL-on-cloud and Spark are common in mid-to-senior DS/analyst job postings.	• freeCodeCamp — Apache Spark / Big Data course • Seattle Data Guy — cloud data engineering concepts	• AWS Skill Builder — cloud fundamentals • Google Cloud Skills Boost — BigQuery free labs • Databricks Academy — free Spark courses
Phase 8 Storytelling & Stakeholder Communication	The most underrated industry skill — translating a finding into a recommendation a non-technical manager can act on. This is what separates a report from a decision.	• Storytelling with Data (Cole Nussbaumer) — official channel • Ken Jee — communicating insights as a DS	• "Storytelling with Data" book + companion course • Coursera — "Data Storytelling" specialization
Phase 9 Portfolio, Capstone & Interview Prep	What actually gets you hired — 2-3 real end-to-end projects (not tutorials), a clean GitHub/portfolio, plus case-study and SQL interview practice.	• Luke Barousse — data analyst project + interview prep • Ken Jee — portfolio project ideas • NeetCode — SQL/DSA interview practice	• StrataScratch — SQL interview questions • LeetCode — SQL/DSA practice • Exponent — DS/analytics interview prep

A Few General Tips

- Alex The Analyst and Luke Barousse are the closest equivalents to Karpathy for the analytics track — both walk through real analyst workflows end-to-end. Don't skip their project videos even if you use paid courses elsewhere.
- The Google Data Analytics Certificate and Andrew Ng's ML Specialization line up almost exactly with Phases 1, 2, and 5 — treat them as your structured backbone and fill in the rest with YouTube.
- Most platform courses above are free to audit — you only pay for a certificate — so use them liberally, especially in Phases 1-5.
- Treat YouTube videos as your primary learning and platforms/courses as structured practice and reinforcement — watching alone won't build the skill.
- Industry hires on projects, not certificates. By the end of Phase 9 you should have 2-3 portfolio projects that go from raw/messy data all the way to a dashboard or written recommendation — that's what recruiters actually look for.

Industry Role Map — Where Each Phase Shows Up on the Job

Role	Core Phases Used Daily	Typical Deliverable
Data Analyst	Phases 1, 2, 3, 6, 8	Dashboards, weekly reports, ad-hoc business queries
Data Scientist	Phases 1, 2, 4, 5, 8	Predictive models, experiment results, recommendations
Analytics Engineer	Phases 1, 2, 7	Clean, reliable data pipelines for others to build on
BI Developer	Phases 3, 6	Self-serve dashboards, KPI tracking systems
ML / Applied Scientist	Phases 1, 4, 5, 7	Production ML models, A/B test frameworks